

FOOD REMEDY API

RE003 – Understanding Calorie Density and Its Impact on Food Choices

Prepared For: Food Remedy API – Research Team

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1. Relationship Between Calories and Portion Size

Calories is the measurement of amount of energy present in the food whereas, Portion size indicates the amount of food eaten. The relationship between calories and portion size is proportional as larger the portion size refers to more calories. Previous conducted research shows that when an individual is served with larger portions, they intake more calories without realising feelings of fullness.

Larger portion of low-calorie density foods provide less calories as compared to small portions of high-density calorie foods, which provide more calories in small portions.

- Comparison Table for Calorie Density for various foods:

Food	Portion	Calories	Density Level
Broccoli	100g	34	Low
Chicken	100g	165	Medium
Chocolate	100g	535	High

2. Examples of Low-Calorie-Density Foods

Most of low-calorie density foods are high in water and fibre, which increases size and volume without adding more calories. Examples of low-calorie-density foods include vegetables and fruits like spinach, lettuce, kale, broccoli, cucumber, watermelon, strawberries, etc.

They have usually more water content and fibre along with low fat content as compared to other high-calorie-density foods. They allow us to consume low calories while intaking less food, which can help with weight-loss.

- Following figure represents some Low-Calorie-Density Foods (fruits and Vegetable)



3. Examples of High-Calorie-Density Foods

High-Calorie-Density Foods contains more fat and sugar content to increase calories per portion. Examples of high-calorie-density foods include Chocolates, candies, chips, fried foods, cheese and fast foods like burger, fries, or pizzas. They are high in calories, so their small portion will contain high amount of calories.

- Following figure represents some High-Calorie-Density Foods (Fast Food)



4. How Calorie Density Influences Diet Decisions

- Many People decide healthiness of food on the basis of portion size instead of calories. Low-calorie-density foods looks healthier to people because less calories will be consumed even after consuming them in large amount.
- Diet which contains low-calorie-density foods can help people to eat more while total intake of calorie. Examples:
 - Choosing fruits over desserts.
 - Adding vegetable to increase volume while maintain calories.
- People do not to completely ignore high calorie foods to maintain healthy weight. Rather they can eat smaller portion of high-density foods and consume them along with low-density foods.
 - Examples: Pizza with large salad or Peanut butter with apple slices.
- Having better understanding of calorie density will help user to interpret food health information by comparing calories present in food per intake or serving, recognizing the specific high-calorie-density food whose small portion can contain many calories and help people to choose food which can provide more volume and nutrients along with less calories.

5. References:

- ScienceDirect (2023). Calorie density and its role in diet and energy intake. Retrieved from <https://www.sciencedirect.com/science/article/pii/S0002916523295398>
- PubMed Central. (2017). Energy density of foods and its impact on energy intake and body weight. Retrieved from <https://pmc.ncbi.nlm.nih.gov/articles/PMC5687574/>